A Representation-Stack Library for Mathematical Structures in Physical Representation

Motives, Periods, Coalgebras, Gauge Redundancy, and Quantum-Informational Availability

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[1]Conceptual mathematics and mathematical physics research note

June 29, 2026

Abstract

We propose a formal framework for a mathematics-to-physics representation library. The organizing thesis is that many physical quantities are not merely described by mathematics, but are obtained as realizations of structured mathematical information. The framework is expressed as a representation stack: abstract mathematical structures are interpreted through realization functors into physical representations and then into observables. The paper organizes algebraic geometry, algebraic topology, category theory, homotopy type theory, sheaf and stack theory, motivic periods, Lie coalgebras, Hopf algebras, and quantum-information error-protection into a single status-labeled dictionary. Standard correspondences, such as topological quantum field theories as functors from cobordism categories, categorical quantum mechanics via dagger-compact categories, gauge fields as local data with descent, and Feynman amplitudes as period-like integrals, are separated from heuristic and speculative ontological extensions. A central guiding formula is

$$\text{observable} = \mathrm{Real}(\text{abstract structure}),$$

while the motivic refinement reads

$$\mathcal{A} = \mathrm{per}(\mathcal{A}^{\mathfrak{m}}), \qquad \Delta(\mathcal{A}^{\mathfrak{m}}) = \sum_i \mathcal{A}^{\mathfrak{m}}_{i,1} \otimes \mathcal{A}^{\mathfrak{m}}_{i,2}.$$

This formalizes the claim that coalgebraic and motivic structures can encode how physical amplitudes, periods, and observables decompose into primitive informational pieces. The result is not presented as an established physical theory, but as an extensible research language for comparing representation, realization, measurement, symmetry redundancy, and logical information across several mature mathematical frameworks.

Keywords: motives; periods; physical representation; stacks; gauge redundancy; algebraic topology; algebraic geometry; HoTT; category theory; Lie coalgebras; Hopf algebras; quantum information.

Contents

1 Introduction

Mathematical physics contains many precise examples in which a physical theory is naturally expressed as a translation from one mathematical category to another. Atiyah's axioms for topological quantum field theory place an n-dimensional TQFT in the form of a symmetric monoidal functor from a bordism category to vector spaces [1]. Lurie's formulation of the cobordism hypothesis refines this idea to extended field theories and higher categories [2]. Categorical quantum mechanics expresses quantum protocols using compact closed and dagger-compact categorical structures [4, 3]. Homotopy type theory interprets identity types as path-like structures and univalence as the principle that equivalent types may be identified [5]. Feynman amplitudes are often periods or period-like integrals and are connected with algebraic geometry, motives, multiple polylogarithms, and Hopf-algebraic renormalization [9, 10, 11, 14, 12].

The present paper assembles these examples into a single representation library. The guiding phrase is:

algebraic structure encoding how physical amplitudes, periods, and observables decompose into primitive informational pieces

We interpret this phrase as a program for organizing mathematical structures by the physical roles they play: composition, decomposition, localization, gauge equivalence, realization, measurement, and informational robustness.

The key move is to distinguish three levels:

$$\textbf{abstract structure} \longrightarrow \textbf{realization} \longrightarrow \textbf{physical representation} \longrightarrow \textbf{observable/measurement}.$$

This is not meant to erase the distinction between mathematics and physics. Rather, it is a language for tracking when a mathematical structure is used as a rigorous model, when it is used as a heuristic analogy, and when it is used as a speculative ontological proposal.

1.1 Contributions

The paper contributes the following conceptual formalization.

1. A *representation-stack* formalism: local mathematical-physical translations form a prestack over domains of mathematics, and become stack-like when compatible local interpretations glue up to equivalence.

2. A status system distinguishing standard mathematical-physics correspondences (S), strong heuristic dictionary entries (H), and speculative ontological extensions (P).

3. A motivic interpretation of "the functions of" an object: not as a replacement for the standard definition of motives, but as a semantic layer in which motives serve as pre-numerical functional cores behind realizations.

4. A coalgebraic anatomy of amplitudes: motivic or period-like amplitudes admit coproduct/coaction/cobracket operations that decompose observables into lower-complexity informational pieces.

5. A stack-theoretic account of gauge redundancy and a quantum-information interpretation of high availability: many local or physical representatives can encode the same logical or physical content.

1.2 Epistemic status

We use the following status labels throughout.

Label	Meaning	Example
S	standard use in mathematics or mathematical physics	TQFT as a functor $\mathrm{Bord}_n \to \mathrm{Vect}$
H	strong heuristic representation dictionary	motive as functional essence
P	speculative philosophical ontology	physical reality as motivic-categorical realization

This separation is crucial. The framework proposed here is intended to be useful precisely because it does not present every analogy as an established theorem.

2 From dictionary to representation stack

2.1 Basic notation

Let **Dom** be a category whose objects are mathematical domains, such as algebraic topology, algebraic geometry, category theory, homotopy type theory, sheaf/stack theory, and motivic period theory. Let each $d \in \mathbf{Dom}$ have an associated category or higher category $\mathbf{Math}_d$ of mathematical structures in that domain. Let **Phys** denote a category, bicategory, or ∞-category of physical representation targets: state spaces, observable algebras, process categories, gauge moduli stacks, quantum code spaces, or measurement outcomes.

The notation

$$[\![M]\!]_{\mathrm{phys}}$$

denotes the physical representation assigned to a mathematical structure M when a chosen interpretation is understood.

Definition 2.1 (Representation entry)**.** *A* representation entry *is a quadruple*

$$E = (M, P, \tau, \sigma),$$

where $M \in \mathbf{Math}_d$ is a mathematical structure, $P \in \mathbf{Phys}$ is a proposed physical representation, τ is a semantic translation datum, and

$$\sigma \in \{\mathsf{S}, \mathsf{H}, \mathsf{P}\}$$

is an epistemic status label.

The semantic translation datum τ may be an actual functor, a natural transformation, an equivalence class of models, a physical interpretation map, or a weaker relation. For example, in standard TQFT the translation datum is a symmetric monoidal functor. In a heuristic dictionary entry, it may be an interpretation rule such as "cohomology class $\mapsto$ stable observable."

Definition 2.2 (Representation library)**.** *A* mathematics-to-physics representation library *is a collection of representation entries closed under the following operations when these operations are defined:*

1. *restriction to subdomains,*
2. *transport along equivalence of mathematical structures,*
3. *composition of compatible translations,*
4. *decomposition through boundary, coproduct, or localization operations,*
5. *realization into observables.*

2.2 The representation stack

The library becomes stack-like when local entries glue, not necessarily by equality, but by equivalence. This captures the phenomenon that local descriptions of a physical system often agree only up to gauge transformation.

Definition 2.3 (Representation prestack)**.** *A* representation prestack *is a pseudofunctor*

$$\mathbf{Rep}_{\mathrm{phys}} : \mathbf{Dom}^{\mathrm{op}} \longrightarrow \mathbf{Gpd},$$

assigning to each domain d a groupoid of representation entries in that domain, and assigning to a domain inclusion or refinement $u : e \to d$ a restriction functor

$$u^* : \mathbf{Rep}_{\mathrm{phys}}(d) \to \mathbf{Rep}_{\mathrm{phys}}(e).$$

Definition 2.4 (Representation stack)**.** *Let* **Dom** *be equipped with a Grothendieck topology expressing domain coverage or local contextual coverage. A representation prestack* $\mathbf{Rep}_{\mathrm{phys}}$ *is a* representation stack *if compatible families of local representation entries glue to global entries, and if the gluing is unique up to a coherent isomorphism.*

Remark 2.1. *The word "stack" is used here in a formal but deliberately broad sense. In ordinary geometry, stacks encode descent with automorphisms. Here, the automorphisms represent gauge changes, duality transformations, code stabilizers, or other equivalences that do not change the relevant physical content.*

2.3 The realization pipeline

The universal representation pipeline is

$$M \in \mathbf{Math}_d \xrightarrow{\Phi} \text{abstract physical-information object} \xrightarrow{\mathrm{Real}_\alpha} \text{physical representation} \xrightarrow{\mathbf{Obs}} \text{measurement data.}$$

Equivalently,

$$\boxed{\mathbf{Obs}_\alpha(M) = \mathbf{Obs}(\mathrm{Real}_\alpha(\Phi(M))).}$$

The index α labels the realization channel: Betti, de Rham, Hodge, etale, p-adic, quantum, thermodynamic, computational, experimental, or operational.

Axiom 2.1 (Realization)**.** *A physical observable is represented as the output of a realization channel applied to an abstract structure:*

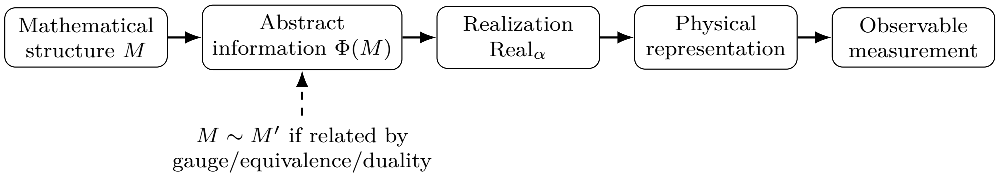
$$\textit{observable} = \mathbf{Obs} \circ \mathrm{Real}_\alpha(\textit{abstract structure}).$$

Axiom 2.2 (Equivalence)**.** *If two mathematical descriptions are related by an equivalence that lies in the automorphism or gauge groupoid of a representation entry, then they determine the same physical content at the chosen level of observation.*

Axiom 2.3 (Locality/descent)**.** *Physical representations must be locally assignable and globally coherent. Failure of descent is interpreted as an obstruction, anomaly, or missing boundary datum.*

Axiom 2.4 (Decomposition)**.** *For compositional or period-like observables, there exists a decomposition operation, such as a boundary, coproduct, coaction, spectral sequence, or factorization rule, that reveals lower-level informational pieces.*

2.4 A master diagram



3 Motives as functional essence

The word "motive" has a precise mathematical history. In Grothendieck's vision, motives should function as universal objects behind cohomology theories. A variety X has several cohomological realizations, and the motive $\mathrm{Mot}(X)$ is a universal source from which these realizations are extracted. This standard picture may be summarized by

$$H_\alpha(X) \simeq \mathrm{Real}_\alpha(\mathrm{Mot}(X)),$$

where α may denote Betti, de Rham, Hodge, etale, or another realization.

The semantic proposal of this paper is not to redefine motives, but to add a representational reading:

$$\boxed{\mathrm{Mot}(X) = \text{the invariant functional essence of } X.}$$

In the language proposed by the user motivating this manuscript, a motive is the representational structure encoding "the functions of" a thing. More precisely, it is the invariant pre-numerical source from which different functional readings of the object can be obtained.

Definition 3.1 (Functional-essence interpretation)**.** *A functional-essence interpretation is a status-H semantic rule*

$$\mathsf{FE}(X) := \mathrm{Mot}(X),$$

read as "the invariant structure from which the functions, cohomological readings, periods, and numerical shadows of X are realized."

This yields the chain

$$\boxed{X \longmapsto \mathrm{Mot}(X) \longmapsto \mathrm{Real}_\alpha(\mathrm{Mot}(X)) \longmapsto \text{physical observable.}}$$

It is often useful to distinguish the motivic level from the numerical level. Let $\mathcal{A}^{\mathfrak{m}}$ be a motivic amplitude object and per a period map. Then

$$\boxed{\mathcal{A} = \mathrm{per}(\mathcal{A}^{\mathfrak{m}})}$$

expresses a physical amplitude as a numerical realization of pre-numerical motivic data.

4 Periods, amplitudes, and coalgebraic anatomy

4.1 Period data

Kontsevich and Zagier describe periods as numbers given by integrals of algebraic differential forms over domains specified by algebraic equations or inequalities [9]. In physical language, a period is a numerical shadow of a geometric pairing:

$$\int_\Gamma \omega.$$

Here ω is a differential form and Γ is an integration cycle or relative chain.

Definition 4.1 (Period datum)**.** *A* period datum *is a tuple*

$$\Pi = (X, D, \omega, \Gamma),$$

where X is an algebraic or analytic space, D is boundary or divisor data, ω is a differential form on $X \setminus D$, and Γ is a cycle or relative chain. Its numerical period is

$$\mathrm{per}(\Pi) = \int_\Gamma \omega.$$

A Feynman integral can often be placed into this pattern after choosing an appropriate representation, regularization, or compactification. Bloch, Esnault, and Kreimer studied motives associated to graph polynomials, motivated by the appearance of multiple zeta values in renormalization data [12]. Brown reviewed the role of periods and motivic periods in Feynman amplitudes [14, 13]. Goncharov developed multiple polylogarithms from analytic, Hodge, and motivic perspectives and described Hopf-algebraic structures on them [11].

4.2 Amplitude objects

Definition 4.2 (Motivic amplitude object)**.** *A* motivic amplitude object *is a pre-numerical object $\mathcal{A}^{\mathfrak{m}}$ in a category or comodule of motivic periods whose numerical realization is a physical amplitude:*

$$\mathrm{per}(\mathcal{A}^{\mathfrak{m}}) = \mathcal{A}.$$

The central decomposition principle is that $\mathcal{A}^{\mathfrak{m}}$ may admit a coproduct or coaction

$$\Delta(\mathcal{A}^{\mathfrak{m}}) = \sum_i \mathcal{A}^{\mathfrak{m}}_{i,1} \otimes \mathcal{A}^{\mathfrak{m}}_{i,2}.$$

This should be interpreted as a full hierarchical anatomy map. A cobracket

$$\delta : \mathcal{L} \longrightarrow \Lambda^2 \mathcal{L}$$

extracts a primitive antisymmetric part of the decomposition.

Definition 4.3 (Coalgebraic anatomy)**.** *Given a class of observable objects $\mathcal{O}$ equipped with a coalgebraic operation Δ or δ, the* coalgebraic anatomy *of an observable $O \in \mathcal{O}$ is the system of lower-complexity pieces obtained by iterating the decomposition operations.*

Proposition 4.1 (Conditional amplitude decomposition)**.** *Suppose motivic amplitude objects form a comodule over a Hopf algebra of motivic periods. Then the coaction defines a representation-stable decomposition of amplitudes into tensorial pieces, and every compatible realization channel transports this decomposition to the corresponding analytic or numerical amplitude representation.*

Proof. This is a formal property of comodules and functorial realization. If Δ is the comodule structure and Real_α is compatible with the relevant tensor product, then applying Real_α to the coaction identity gives a corresponding decomposition in the target category. The physical content of the statement depends on whether the amplitudes under consideration actually admit such motivic lifts and compatible realizations. $\square$

4.3 Coproducts, cuts, and factorization

The dictionary proposed here reads:

$$\Delta = \text{full anatomy map}, \qquad \delta = \text{primitive information split}.$$

In amplitude language, such decompositions may be compared with cuts, factorization channels, branch discontinuities, residues, and boundary strata of positive geometries. Positive geometries and canonical forms provide one rigorous setting in which boundaries and residues are linked to scattering-amplitude structures [16].

5 Algebraic geometry: spaces of physical possibility

Algebraic geometry contributes the language of solution spaces, moduli, singularities, compactifications, sheaves, stacks, periods, and arithmetic refinements. The core representation principle is

$$\boxed{\text{algebraic geometry} = \text{geometry of physical possibilities.}}$$

A coordinate ring is an algebra of functions; physically, this suggests that observables generate the space. A quotient ring R/I represents functions modulo constraints, which can be read as the on-shell observable algebra. A moduli stack retains automorphisms and therefore models gauge-correct configuration spaces.

Example 5.1 (On-shell observable algebra)**.** *Let R be an algebra of fields or coordinates and I the ideal generated by equations of motion or constraints. Then R/I is a natural algebraic representation of observables modulo the imposed physical equations. This entry is standard in algebraic approaches to field theory, but its ontological reading as "measurements generate space" is heuristic.*

Example 5.2 (Positive geometry)**.** *A positive geometry is a geometric region equipped with a canonical form whose boundary residues recursively encode lower-dimensional canonical forms. In scattering-amplitude applications, this gives a precise model of how geometric boundaries can represent physical singularities or factorization channels [16].*

6 Algebraic topology: conserved global information

Algebraic topology contributes invariants that survive continuous deformation: homotopy classes, homology, cohomology, characteristic classes, generalized cohomology, K-theory, and bordism. The core representation principle is

$$\boxed{\text{algebraic topology} = \text{grammar of conserved global information.}}$$

A cohomology class is a stable observable insensitive to local deformation. Characteristic classes encode quantized topological information in bundles. Spin structures control the possibility of globally defining fermions. Bordism groups and generalized cohomology theories classify topological phases and anomalies in many modern settings. Dijkgraaf-Witten theory gives a concrete example in which group cohomology defines topological gauge theories for finite groups [7].

7 Category theory and HoTT: composition and identity

Category theory turns attention from objects to morphisms, composition, functorial translation, and universal properties. The representation principle is

$$\boxed{\text{category theory} = \text{grammar of physical composition.}}$$

A physical theory may be a functor, a duality may be an equivalence of categories, and a process may be a morphism. In TQFT, the functorial slogan is

$$Z : \mathrm{Bord}_n \longrightarrow \mathcal{C},$$

where boundaries are assigned state objects and cobordisms are assigned processes [1, 2].

Homotopy type theory contributes an internal logic of spaces, paths, equivalences, and higher identities. The representation principle is

$$\boxed{\text{HoTT} = \text{logic of spaces, paths, equivalences, and higher identity.}}$$

A type A may be read as a space of possibilities, a term $a : A$ as an actual state or witness, an identity type $a = b$ as a path or equivalence, and a dependent type $B(x)$ as a field of data over a base configuration. The univalence principle supports the representation-theoretic slogan

$$\boxed{\text{equivalent representations can be treated as the same physical content.}}$$

Cohesive and differential refinements of higher topos theory and type theory further connect these ideas with gauge fields, differential cohomology, Chern-Weil theory, and higher geometric structures [6].

8 Sheaves, stacks, gauge redundancy, and quantum high availability

8.1 Symmetry redundancy

Symmetry redundancy means that different mathematical descriptions represent the same physical state. Gauge theory is the archetypal example. In electromagnetism, the gauge potential A may change by

$$A \mapsto A + d\lambda,$$

while the field strength

$$F = dA$$

is unchanged because $d^2 = 0$. The physical state is not the raw potential; it is the equivalence class of potentials modulo gauge transformation, together with possible global and automorphism data.

Definition 8.1 (Gauge redundancy)**.** *Let X be a space of mathematical descriptions and G a group or groupoid acting on X. A* gauge redundancy *is an action for which points in the same orbit represent the same physical content at the chosen observational level.*

The coarse quotient X/G identifies equivalent descriptions, but the quotient stack $[X/G]$ retains stabilizers:

$$\mathrm{Aut}_{[X/G]}(x) \simeq \mathrm{Stab}_G(x).$$

Thus a stack is not merely a quotient; it remembers how descriptions are equivalent.

Proposition 8.1 (Stacks retain gauge information)**.** *If a physical configuration space is represented by descriptions X modulo gauge transformations G, and if stabilizer groups influence observables, anomalies, quantization, or path-integral weights, then the quotient stack $[X/G]$ is a more faithful representation object than the coarse quotient X/G.*

Proof. The coarse quotient records orbits but generally forgets automorphism groups of representatives. The stack quotient records objects, equivalences, and self-equivalences. Gauge-theoretic data such as residual symmetry, isotropy, and automorphism factors live in these self-equivalences, so retaining the stack structure preserves information discarded by the coarse quotient. □

8.2 Quantum-information high availability

In classical high availability, information is often protected by replication. In quantum information, arbitrary quantum states cannot simply be copied; robust storage is achieved by encoding logical information into a larger physical Hilbert space. Topological codes and anyonic models are standard examples of fault-tolerant encodings in which information is protected by global or topological structure [8].

The representation-stack viewpoint suggests the following dictionary:

$$\boxed{\text{many physical representatives} \longrightarrow \text{one logical/observable content.}}$$

A stabilizer quantum code has operators s satisfying

$$s|\psi_L\rangle = |\psi_L\rangle$$

for logical code states $|\psi_L\rangle$. In a subsystem code, the physical Hilbert space may decompose schematically as

$$\mathcal{H}_{\text{phys}} \simeq \mathcal{H}_{\text{logical}} \otimes \mathcal{H}_{\text{gauge}} \oplus \mathcal{H}_{\text{error}},$$

where changes in the gauge subsystem do not change the encoded logical content. This is the quantum-information cousin of gauge redundancy.

Definition 8.2 (Quantum high-availability representation)**.** *A* quantum high-availability representation *is an encoding*

$$\iota : \mathcal{H}_{\text{logical}} \hookrightarrow \mathcal{H}_{\text{phys}}$$

together with an equivalence groupoid of physical variations such that logical observables are invariant under the allowed variations and detectable/correctable errors move the system within controlled equivalence or syndrome sectors.

Remark 8.1. *The analogy with stacks is structural: a sheaf stores compatible local data; a stack stores compatible local data plus automorphisms; a quantum code stores logical data across physical degrees of freedom plus transformations that do not change the logical content. This is not a claim that every quantum code is literally a stack, but it suggests a useful representation language for fault tolerance, gauge subsystems, and topological order.*

9 Axioms for a representation ontology

The following axioms summarize the proposed research program.

Axiom 9.1 (Syntax)**.** *Mathematical structures supply the syntax of representation: objects, morphisms, spaces, paths, functions, forms, cycles, symmetries, and decompositions.*

Axiom 9.2 (Semantics)**.** *A physical interpretation is a semantic assignment from mathematical structures to physical roles. This assignment must record its epistemic status: standard, heuristic, or speculative.*

Axiom 9.3 (Realization)**.** *Measurements are obtained only after choosing a realization channel. Different realization channels can expose different physical readings of the same abstract structure.*

Axiom 9.4 (Equivalence/gauge)**.** *Physical content is invariant under designated equivalences. The correct representation object should retain the automorphisms of these equivalences when they affect physical or informational behavior.*

Axiom 9.5 (Decomposition)**.** *Composite observables should be studied through decomposition operations: boundary, residue, coproduct, coaction, spectral sequence, localization, or factorization.*

Axiom 9.6 (Motivic pre-numericality)**.** *When physical observables are periods or period-like quantities, their numerical values should be distinguished from the pre-numerical structures whose realizations produce them.*

Conjecture 9.1 (Motivic-categorical information ontology)**.** *A sufficiently broad class of physical theories admits a representation-stack description in which states, processes, observables, amplitudes, gauge redundancies, and measurement values arise as realizations of structured categorical, homotopical, geometric, and motivic information.*

The conjecture is intentionally broad. Its value lies in generating testable subprograms: identify the domain, define the representation entries, specify the realization channels, distinguish equivalences from physical transformations, and compute decomposition operations.

10 Limitations and criteria for use

The framework is not a replacement for physical modeling, experiment, or established mathematical definitions. Several cautions are necessary.

1. Not every mathematical analogy is a physical theory. The status labels are part of the formalism.
2. Motives are not literally defined as "the functions of" an object in standard mathematics; that phrase is a semantic interpretation proposed for this library.
3. A single universal functor from all mathematics to all physics is unlikely to be well-defined. The representation-stack approach is deliberately local and domain-indexed.
4. Some physical observables are not known to be periods, and some amplitudes require elliptic, modular, non-Tate, or more general analytic structures.
5. Gauge redundancy is not the same as physical symmetry. Physical symmetries can map one physical state to another; gauge symmetries identify different descriptions of the same physical content.

11 Conclusion

The proposed library can be compressed into four slogans:

$$\boxed{\text{mathematics is the syntax of physical representation;}}$$

$$\boxed{\text{motives are functional semantics;}}$$

$$\boxed{\text{periods are numerical measurements;}}$$

$$\boxed{\text{coalgebras are decomposition laws.}}$$

More formally, a physical observable is modeled as a realization of an abstract mathematical object, and a decomposable observable is modeled by equipping that object with boundary, coproduct, coaction, or cobracket structure. Algebraic topology tracks conserved global information. Algebraic geometry tracks spaces of possibility and singularities. Category theory tracks composition and translation. HoTT tracks identity and equivalence. Sheaves and stacks track local-to-global information with gauge redundancy. Motives track pre-numerical functional essence. Periods track measured numerical shadows. Lie coalgebras and Hopf algebras track the anatomy of amplitudes, renormalization, and primitive informational pieces.

The resulting framework is not the final ontology of physics. It is a disciplined language for building one.

A Core translation library

Mathematics	Physical representation	Semantic interpretation	Status
Mathematical object X	Physical system or possible world-structure	Structured carrier of physical information	H
Representation $\rho : X \to \mathrm{End}(V)$	Physical realization on states	How abstract structure acts on observables or states	S
Realization functor	Translation into measurable form	Passage from abstract information to physically readable data	S/H
Invariant	Conservation law or observable quantity	What remains stable under allowed transformations	S
Equivalence	Same physics in different mathematical coordinates	Dual descriptions, gauge equivalence, change of representation	S
Symmetry	Redundancy or physical transformation	Law preserving structure	S
Duality	Two theories with same physical content	Equivalence between representation languages	S
Moduli space	Space of physically distinct configurations	Parameter space of possible structures	S
Quotient	Identification of redundant descriptions	Gauge reduction or physical state space	S
Boundary	Physical interface or asymptotic region	Where factorization, measurement, or interaction appears	S/H
Singularity	Threshold, phase transition, divergent region	Breakdown or special physical regime	S/H

Mathematics	Physical representation	Semantic interpretation	Status
Decomposition	Factorization, cut, measurement split	Observable broken into primitive pieces	S
Primitive	Irreducible informational unit	Cannot be decomposed further inside the chosen grammar	H
Functor	Physical theory as translation	Assigns states to spaces and processes to histories	S
Natural transformation	Equivalence or deformation of theories	Consistent translation between physical descriptions	S/H
Category	Universe of systems and processes	Objects are systems; morphisms are allowed transitions	S/H

B Motivic and amplitude library

Mathematics	Physical representation	Decomposition meaning	Status
Field F	Kinematic or coordinate domain	Allowed values of physical parameters	S/H
$F^{\times}$	Nonzero scales or ratios	Energies, masses, cross-ratios, coupling ratios	S/H
Cross-ratio	Conformal invariant	Dimensionless observable independent of scale	S
Algebraic variety	Constraint solution space	Allowed configurations satisfying equations	S
Graph hypersurface	Geometry associated to Feynman graph	Algebraic container of graph integral data	S
Feynman integral	Period integral	Observable from pairing geometry and domain	S
Period	Measurable number from geometry	Numerical shadow of deeper structure	S
Period pairing $\int_{\Gamma} \omega$	Observable amplitude contribution	Pairing of form and integration cycle	S
Differential form ω	Integrand or local observable density	What is being accumulated	S
Cycle Γ	Integration domain, path, history class	Where physical accumulation occurs	S
Relative cohomology	Boundary-sensitive observable	Observable with cuts, thresholds, or boundaries	S
Motive	Pre-numerical functional essence	Hidden source of periods	S/H
Mixed motive	Layered informational object	Amplitude with multiple weights and depths	S/H
Mixed Tate motive	Polylogarithmic period structure	Tame amplitude period class	S
Non-Tate motive	More complex amplitude geometry	Elliptic, Calabi-Yau, or modular sectors	S
Multiple zeta value	Special period	Constant term in loop or amplitude expansions	S
Polylogarithm $\mathrm{Li}_n(x)$	Layered propagation function	Iterated amplitude contribution	S

Mathematics	Physical representation	Decomposition meaning	Status
Multiple polylogarithm	Multi-scale iterated propagation	Nested channel dependence	S
Symbol of polylogarithm	First-order function decomposition	Branch-cut or discontinuity data	S
Coproduct/coaction	Decomposition of period	How amplitude splits into lower pieces	S
Cobracket δ	Antisymmetric primitive split	Factorization channel or information split	H
Goncharov coproduct	Motivic decomposition operation	Arithmetic anatomy of polylogs	S
Goncharov Lie coalgebra	Coalgebra of polylog motivic data	Grammar of primitive amplitude data	S/H
Weight grading	Transcendental complexity	Loop depth or functional complexity	S/H
Depth grading	Iterated-integral nesting	Number of nested propagation layers	S/H
Primitive element	Indecomposable motivic component	Irreducible informational contribution	H
Regulator map	Motive to analytic number	Measurement or numerical extraction	S/H
Hodge realization	Analytic-geometric reading	Differential-form representation	S
de Rham realization	Form/integrand side	Calculational analytic representation	S
Betti realization	Cycle/topological side	Path or integration-domain representation	S
Etale realization	Arithmetic/discrete realization	Number-theoretic shadow	S/H
Cosmic Galois group	Symmetry acting on periods	Hidden symmetry organizing amplitudes	H/P

C Algebraic geometry library

Algebraic geometry	Physical representation	Semantic meaning	Status
Affine variety	Classical solution space	Allowed states satisfying polynomial constraints	S
Projective variety	Scale-invariant configuration space	Physical data modulo rescaling	S
Scheme	Generalized solution space	Includes infinitesimal or nilpotent structure	S/H
Spec R	Space encoded by observables	Observables determine geometry	S/H
Coordinate ring	Algebra of observables/functions	Measurements generate space	S
Ideal I	Constraint equations	Equations of motion or conservation constraints	S/H
Quotient ring R/I	On-shell observable algebra	Functions modulo physical constraints	S
Nilpotent element	Infinitesimal thickening	Virtual, off-shell, or ghost-like direction	H
Sheaf	Locally defined data	Local fields glued into global fields	S
Stalk	Local data at a point	Infinitesimal local physical information	S/H
Cech cocycle	Gluing rule	Gauge transition function	S
Sheaf cohomology	Obstruction to global gluing	Global anomaly or obstruction	S/H
Line bundle	Phase/gauge degree over space	$U(1)$-type local phase structure	S
Vector bundle	Internal degrees over spacetime	Matter or gauge field carrier	S
Principal G-bundle	Gauge configuration	Local symmetry fiber at each point	S
Connection	Gauge potential	Rule for parallel transport	S

Algebraic geometry	Physical representation	Semantic meaning	Status
Curvature	Field strength	Failure of transport to commute	S
Divisor	Codimension-one locus	Defect, pole, boundary, charge surface	H
Blowup	Resolution of singularity	Regularization or zooming into divergence	S/H
Exceptional divisor	New boundary after resolution	Counterterm or hidden boundary sector	H
Compactification	Add boundary at infinity	Include asymptotic states or IR sectors	S/H
Moduli space	Space of inequivalent structures	Physically distinct vacua or solutions	S
Moduli stack	Moduli with automorphisms retained	Gauge quotient preserving stabilizers	S
Quotient stack $[X/G]$	Gauge quotient	Physical states modulo symmetry, with isotropy	S
Derived scheme/stack	Equations plus higher constraints	BV/BRST-style field space with ghosts	S/H
Derived critical locus	Critical points with obstruction data	Perturbative field theory around action	S/H
Tangent complex	Linearized deformation data	Fluctuations around solution	S
Cotangent complex	Moment/covector deformation data	Phase-space infinitesimals	S/H
Symplectic variety	Geometric phase space	Classical mechanics structure	S
Poisson variety	Bracketed observable space	Classical observable algebra	S
Calabi-Yau variety	Special geometry or compactification	String background or higher period source	S
Elliptic curve	Torus-like period geometry	Elliptic Feynman integral sector	S
Algebraic curve	Spectral curve	Dispersion or integrability data	S
Jacobian	Torus of line bundles	Abelianized phase/integrable variables	S
Picard group	Line-bundle classes	Phase sectors or charge classes	S/H
Hodge structure	Cohomological decomposition	Analytic-topological complexity of observables	S
Mixed Hodge structure	Layered singular/relative cohomology	Amplitudes with boundaries and degenerations	S
Variation of Hodge structure	Periods varying over parameters	Amplitudes over kinematic space	S
Gauss-Manin connection	Transport of cohomology over moduli	Differential equations for amplitudes	S
Picard-Fuchs equation	Period differential equation	Equation satisfied by integral or amplitude	S
Riemann-Hilbert correspondence	Differential equations to monodromy	Dynamics to analytic-continuation data	S
Monodromy	Transport around singularity	Branch cuts or analytic continuation	S
Discriminant locus	Degenerate parameter values	Thresholds or singular kinematics	S
Positive geometry	Region with canonical form	Geometry encoding scattering amplitude	S
Canonical form	Distinguished differential form	Amplitude or integrand form	S
Amplituhedron	Positive geometry for amplitudes	Scattering encoded geometrically	S
Positive Grassmannian	Positive cell geometry	On-shell amplitude combinatorics	S
Cluster algebra	Mutating coordinate system	Different amplitude charts/channels	S/H
Cluster coordinate	Natural amplitude variable	Symbol letter or physical cross-ratio	S

Algebraic geometry	Physical representation	Semantic meaning	Status
Tropical geometry	Degeneration/asymptotic geometry	Classical limit, graph limit, boundary regime	S/H
Toric variety	Combinatorial algebraic geometry	Phase space from lattice/polytope data	S
Newton polytope	Monomial support geometry	Semiclassical or dominant-balance structure	H

D Algebraic topology library

Algebraic topology	Physical representation	Semantic meaning	Status
Topological space X	Configuration/spacetime/state space	Domain of physical possibility	S
Point	State or event	Localized physical datum	S
Path	Evolution/worldline/process	Continuous transition	S
Loop	Closed evolution	Recurrence or holonomy probe	S
Homotopy	Continuous deformation	Physically equivalent path/process	S
Homotopy class	Topological sector	Deformation-invariant physical class	S
Fundamental group $\pi_1(X)$	Loop/winding structure	Monodromy, holonomy, defect winding	S
Higher homotopy $\pi_n(X)$	Higher-dimensional holes	Brane or defect charge sectors	S/H
Homology $H_n(X)$	Cycles or holes	Conserved extended structures	S
Cohomology $H^n(X)$	Fields or flux observables	Functions on cycles or charge detectors	S
Chain	Formal region	Piece of physical domain/history	S
Boundary operator ∂	Boundary of region	Where conservation or factorization appears	S
Cycle $\partial c = 0$	Closed conserved object	No boundary; conserved charge carrier	S
Boundary $c = \partial b$	Trivial conserved object	Pure gauge or physically null sector	H
Cocycle	Closed field or charge rule	Conservation condition	S
Coboundary	Gauge-exact field	Redundant potential contribution	S
Exact sequence	Constraint propagation	Conservation and obstruction bookkeeping	S
Long exact sequence	Boundary-bulk relation	Defect or boundary charge transfer	S
Mayer-Vietoris sequence	Gluing local regions	Reconstruct global physics from patches	S
Cup product	Product of cohomology classes	Interaction of observables or charges	S/H
Cap product	Pairing class with cycle	Measurement of charge over region	S
Poincare duality	Cycles to fields	Defects as dual field strengths	S
de Rham cohomology	Differential forms modulo exact forms	Gauge-invariant flux data	S
Differential form	Local field density	Quantity integrated over regions	S
Closed form	Conserved field	$d\omega = 0$ conservation law	S
Exact form	Gauge-trivial field	$\omega = d\alpha$ redundancy	S
Hodge star $\star$	Dual field operation	Electromagnetic or geometric duality	S
Fiber bundle	Local degrees over base	Field with internal state at each point	S
Principal bundle	Gauge field topology	Gauge choices glued globally	S
Classifying space BG	Universal gauge-bundle classifier	Space of possible G-gauge sectors	S

Algebraic topology	Physical representation	Semantic meaning	Status
Characteristic class	Topological invariant of bundle	Quantized charge, anomaly, or response	S
Chern class	Complex bundle invariant	Monopole, quantized flux, topological-band data	S
Chern character	Cohomological charge map	Topological response or index data	S
Stiefel-Whitney class	Real bundle obstruction	Orientation or spin obstruction	S
Spin structure	Lift of frame data	Fermion consistency condition	S
Index theorem	Topology to analysis	Zero modes or anomalies from topology	S
K-theory	Generalized cohomology of bundles	D-brane charges or topological phases	S
Spectra	Stable homotopy objects	Generalized charge theories	S
Generalized cohomology	Beyond ordinary charge theory	Exotic charges, phases, anomalies	S
Steenrod operations	Cohomology operations	Constraints on possible phases or anomalies	S/H
Postnikov tower	Layered homotopy data	Hierarchy of defects or symmetries	H
Fibration	Parameterized family of spaces	Field over configuration or constrained system	S
Long exact homotopy sequence	Fiber/base/total relation	Constraint propagation through gauge structure	S
Loop space ΩX	Space of closed paths	Dynamics of recurrence or based excitations	S/H
Suspension ΣX	Dimension-shifted space	Dimensional lift of defect data	H
Cobordism	Manifold as transition	Spacetime history between boundaries	S
Bordism group	Manifolds modulo boundary	Classification of phases/anomalies	S
TQFT	Functor from cobordisms to vector spaces/categories	Physics as topology-to-algebra translation	S
Boundary TQFT	Theory with interfaces	Edge modes or boundary observables	S
Extended TQFT	Assigns data to all codimensions	Particles, lines, surfaces, defects	S
Fully dualizable object	Point-data determining TQFT	Local seed of global topological theory	S
Group cohomology	Cohomology of symmetry group	Discrete topological action or SPT data	S
Dijkgraaf-Witten theory	Finite-group topological gauge theory	Topological action from group cocycle	S

E Category theory and HoTT library

Mathematics	Physical representation	Semantic meaning	Status
Object	Physical system/state space	Thing capable of processes	S/H
Morphism $f : A \to B$	Process, transition, evolution	Physical transformation	S
Composition $g \circ f$	Sequential process	Do one process after another	S
Identity morphism	Do-nothing process	Persistence or trivial evolution	S
Isomorphism	Reversible equivalence	Same structure in different coordinates	S
Equivalence of categories	Same theory structurally	Dual physical descriptions	S
Functor	Translation between worlds	Physical theory, quantization, representation	S
Natural transformation	Uniform deformation of translations	Theory equivalence or field redefinition	S/H

Mathematics	Physical representation	Semantic meaning	Status
Category of states	State-system universe	Objects are systems; arrows are evolutions	H
Category of observables	Measurement algebra universe	Arrows are observable transformations	H
Monoidal category	Parallel composition	Composite systems	S
Tensor product $\otimes$	Combine systems	Parallel physical composition	S
Unit object I	Vacuum/trivial system	No-system baseline	S/H
Symmetric monoidal category	Exchangeable systems	Bosonic-like swap symmetry	S/H
Braided monoidal category	Nontrivial exchange	Anyonic or braided statistics	S
Ribbon category	Braiding plus twist	Topological quantum computation or knot invariants	S
Compact closed category	Inputs/outputs dualizable	Process-state duality	S
Dagger category	Reversal/adjoint	Quantum adjoint, time reversal, test-state dual	S
Dagger-compact category	Quantum process calculus	Teleportation, entanglement, cups/caps	S
String diagram	Diagrammatic process expression	Circuit or Feynman-like composition grammar	S
Trace	Feedback loop	Closed process, partition function, loop	S/H
Dual object A^*	Antisystem or dual state space	Bra/ket or particle/antiparticle-like relation	S/H
Evaluation map	Pairing	Measurement, annihilation, contraction	S/H
Coevaluation map	Pair creation	Entangled pair or vacuum insertion	S/H
Limit	Universal compatible solution	Constraint satisfaction	S
Colimit	Universal gluing	Build global system from parts	S
Pullback	Fibered constraint	Simultaneous satisfaction of conditions	S
Pushout	Gluing along common boundary	Coupled systems or boundary identification	S
Adjunction $F \dashv G$	Dual translation pair	Free/forgetful, coarse/fine, preparation/measurement	S/H
Monad	Dynamics with context/effects	State update, probability, computation effect	S/H
Comonad	Context extraction	Observation or coarse-grained environment	H
Algebra over monad	System obeying effect structure	Dynamics closed under update	H
Coalgebra	State-transition/decomposition system	Observable unfolding or branching behavior	S/H
Operad	Multi-input composition law	Interaction vertices or allowed couplings	S
PROP	Network of multi-input/output operations	Circuit or Feynman process algebra	S
Enriched category	Structured morphism spaces	Distances, probabilities, Hilbert spaces of processes	S/H
2-category	Objects, processes, transformations	Theories, defects, transformations	S
∞-category	All higher coherences	Histories, homotopies, higher gauge data	S
Higher category	Nested process hierarchy	Particles, strings, branes, defects	S
n-category	n-level process system	Extended TQFT codimension hierarchy	S
Topos	Universe of variable sets/logics	Contextual world of observation	H

Mathematics	Physical representation	Semantic meaning	Status
Sheaf topos	Local-to-global universe	Observer-dependent gluing of information	S/H
Internal language	Logic of a category	Native reasoning system of a physical universe	H
Yoneda lemma	Object known by probes	System determined by all possible observations	S/H
Type A	Space of states, propositions, systems	Domain of possible inhabitants	S/H
Term $a : A$	State, event, witness	Actual instance of a possibility	S/H
Dependent type $B(x)$	Fiber over configuration	Field degrees depending on base point	S/H
Σ-type	Total space	Configuration plus dependent data	S
Π-type	Space of sections/functions	Field assigning data at every point	S
Identity type $a = b$	Path from a to b	Equivalence or evolution between states	S/H
Higher identity type	Homotopy between paths	Coherence between evolutions	S/H
Path induction	Reasoning from identity	Transport laws generated by equivalence	S/H
Transport	Move structure along path	Parallel transport or coordinate change	S/H
Equivalence $A \simeq B$	Same physics in different representation	Duality or gauge-equivalent description	H
Univalence	Equivalence can be treated as identity	Physically equivalent systems identified	S/H
Universe $\mathcal{U}$	Type of systems/theories	Meta-space of physical descriptions	H
Higher inductive type	Generated points, paths, cells	Build topological spaces synthetically	S/H
Circle HIT S^1	Generated loop space	Winding sector or phase circle	H
Sphere HIT S^n	Generated n-charge carrier	Higher topological charge object	H
Pushout type	Glued system	Boundary coupling or fusion of spaces	S/H
Pullback type	Constraint-matched system	Simultaneous compatibility condition	S/H
Quotient type	Identified states	Gauge quotient or equivalence classes	H
Truncation	Forget higher identity data	Classical approximation or coarse-graining	H
Mere proposition	Yes/no observable	Proof-irrelevant physical fact	H
Set-truncation	Classical state set	No higher path information retained	H
Groupoid level	Gauge-state structure	States plus equivalences	S/H
∞-groupoid	Full homotopy state space	States, paths, paths-between-paths, etc.	S/H
Contractible type	Unique state up to equivalence	Trivial phase or vacuum sector	H
Fiber sequence	Structured dependency	Gauge fiber over base configuration	S/H
Modal type theory	Observation/filtering modes	Coarse-graining, localization, discrete views	H
Cohesive HoTT	Synthetic smooth/topological geometry	Internal language for spacetime/gauge fields	S/H
Shape modality	Underlying homotopy type	Forget geometry, keep topology	S/H
Flat modality	Discrete/codified form	Locally constant/arithmetic representation	H
Sharp modality	Codiscrete/crisp form	Purely formal or global representation	H
Differential cohesion	Smooth infinitesimal structure	Synthetic differential geometry for fields	S/H

F Lie, Hopf, sheaf, stack, and decomposition library

Mathematics	Physical representation	Semantic meaning	Status
Group G	Symmetry group	Allowed transformations	S
Lie group	Continuous symmetry	Smooth physical transformations	S
Lie algebra $\mathfrak{g}$	Infinitesimal symmetry	Generators of transformations	S
Bracket $[X, Y]$	Noncommutativity of generators	Curvature or interaction of transformations	S
Representation of Lie algebra	Action on states	How symmetries act physically	S
Universal enveloping algebra	Algebra of symmetry operations	Composite infinitesimal transformations	S
Coalgebra	Decomposition structure	How objects split into observable pieces	S/H
Lie coalgebra	Primitive antisymmetric decomposition	Infinitesimal factorization grammar	H
Cobracket δ	Decomposition map	Cut, channel, or boundary split	H
Bialgebra	Composition plus decomposition	Build and analyze processes	S/H
Hopf algebra	Algebra with antipode	Renormalization, subtraction, recursive cancellation	S
Coproduct Δ	Full decomposition	Splitting graph/function/object into subpieces	S
Antipode S	Algebraic subtraction/inversion	Counterterm or renormalization inverse	S/H
Primitive element	Only trivial coproduct pieces	Irreducible contribution	S/H
Character	Algebra map to numbers/functions	Feynman rule or evaluation rule	S
Connes-Kreimer Hopf algebra	Hopf algebra of Feynman graphs	Nested subdivergence bookkeeping	S
Rooted tree	Divergence hierarchy	Nested renormalization structure	S
Goncharov Lie coalgebra	Coalgebra of polylog motivic data	Primitive period/amplitude decomposition	S/H
Bloch group	Dilogarithmic motivic object	Weight-2 amplitude/volume structure	S/H
K-theory of field	Deep arithmetic invariant	Hidden charge or period structure	S/H
Shuffle relation	Product relation of iterated integrals	Equivalent propagation orderings	S
Stuffle relation	Series multiplication relation	Discrete sum-channel decomposition	S/H
Coaction principle	Periods decompose into tensors	Amplitude space stable under decomposition	S/H
Weight filtration	Complexity hierarchy	Transcendental or loop-depth hierarchy	S/H
Presheaf	Assign data to regions	Local observable assignment	S
Sheaf	Compatible local data glue globally	Physical fields from local measurements	S
Cosheaf	Local data assemble covariantly	Flows, networks, distributed systems	S/H
Stalk	Infinitesimal local data	Field germ at event	S
Section	Choice of data over region	Field configuration	S
Global section	Globally valid field	Full physical configuration	S
Obstruction to section	Failure of global consistency	Anomaly or topological obstruction	S/H
Descent	Reconstruct global from local	Coordinate-independent physics	S
Stack	Sheaf with automorphisms	Gauge fields with symmetry redundancy	S

Mathematics	Physical representation	Semantic meaning	Status
Higher stack	Stack with higher morphisms	Higher gauge fields or branes	S
Moduli stack	Space of solutions with symmetries	Gauge-correct configuration space	S
Classifying stack BG	Universal G-bundle object	Gauge field classifier	S
Gerbe	Higher bundle	B-field or higher gauge potential	S
n-gerbe	Higher gauge object	p-form gauge field carrier	S
Descent cocycle	Transition data	Gauge patching rules	S
Stack quotient	Gauge quotient with stabilizers	Physical quotient retaining symmetry data	S

G Operations and physical interpretations

Mathematical operation	Physical representation	Interpretation
Boundary ∂	Edge of physical region	Where flux or factorization appears
Differential d	Infinitesimal change	Local variation
Coboundary d^*	Gauge/exact shift	Redundant change
Cobracket δ	Primitive split	Irreducible factorization
Coproduct Δ	Full split	Complete hierarchical decomposition
Residue	Boundary contribution	Pole or singular channel
Monodromy	Loop around singularity	Analytic continuation or branch behavior
Filtration	Layered approximation	Scale or complexity hierarchy
Grading	Complexity label	Charge, degree, loop, or weight
Spectral sequence	Successive approximation	Multi-scale obstruction computation
Localization	Focus near region/prime	Effective physics at scale or sector
Completion	Formal neighborhood	Perturbation expansion
Blowup	Resolve singularity	Regularize divergence
Quotient	Identify redundancy	Gauge reduction
Pullback	Enforce compatibility	Constraint matching
Pushout	Glue along boundary	Couple systems
Trace	Close process loop	Partition function or loop amplitude
Dual	Reverse/probe object	Measurement, conjugate, anti-object
Tensor product	Combine independent systems	Parallel composition
Direct sum	Alternative sectors	Superselection or additive alternatives
Hom	Space of transformations	Possible processes between systems
Ext	Obstruction or extension class	Bound state, anomaly, deformation
Tor	Derived intersection correction	Hidden compatibility failure

H Mathematical domains as physical faculties

Domain	What it represents physically
Algebra	Composition of observables and transformations
Coalgebra	Decomposition of observables and processes

Domain	What it represents physically
Topology	Global invariants under continuous deformation
Algebraic topology	Conserved charges, defects, anomalies, sectors
Differential geometry	Smooth fields, curvature, spacetime dynamics
Algebraic geometry	Solution spaces, moduli, singularities, periods
Arithmetic geometry	Hidden number-theoretic structure of observables
Category theory	Composition, translation, systems, processes
Higher category theory	Defects, interfaces, higher processes
HoTT	Internal logic of spaces and equivalences
Sheaf theory	Local-to-global physical information
Stack theory	Gauge-correct local-to-global information
Motive theory	Universal functional essence behind realizations
Homological algebra	Obstructions, resolutions, derived consistency
K-theory	Stable charge and vector-bundle classification
Operator algebra	Quantum observables and noncommutative spaces
Representation theory	How symmetry becomes physical action
Symplectic geometry	Classical phase space
Poisson geometry	Bracket structure of observables
Noncommutative geometry	Geometry encoded by noncommuting observables
Tropical geometry	Degeneration, asymptotics, combinatorial limit
Cluster algebra	Mutating coordinate systems for amplitudes
Operad theory	Interaction vertices and compositional laws

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